

AI for Economic Research: Dynamic Models, Language, and Agents

Lecture 10.1: Case Studies — Framing

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Topic 10.1 Agenda

1. **Framing** — why case studies, what they show, road map for the lecture

Framing

From Lec. 9 concepts to complete research workflows

Where We Are in the Course

1–6: models + AI/ML tools

7–8: LLMs + RAG

9: agentic AI workflow

10: case studies

- Lec. 9 gave the workflow: specification, structure, validation, and human oversight
- Lec. 10 shows how that workflow changes shape across different research jobs
- The core question is no longer “what is a skill?” It is “what workflow should I build for this task?”

Lab Readiness Check

Before today's exercises, verify all five:

1. **Terminal agent ready** — one of Claude Code, Codex, Gemini CLI, Cursor, or a course-approved equivalent can run
2. **Access confirmed** — login, quota, and model access checked before class
3. **Git initialized in your project** — `git status` runs without error
4. **Working Python or R environment** — `which python` or `which R` returns a path
5. **Project memory created** — `AGENTS.md`, `CLAUDE.md`, `CODEX.md`, or equivalent rules file with project name, language, and goal

If any box is unchecked, open Appendix A now. The exercises in this lecture require all five. If setup takes more than 10 minutes, pair with a neighbor and share one working project.

Why Case Studies After Lec. 9

- Knowing the vocabulary of agentic AI does not tell you how to design a research workflow
- The difficult step is decomposition: where to split the task, where to validate, and what stays human
- Different tasks need different harnesses: paper review is not the same as MEPS harmonization, AI-exposure measurement, OLG homotopy, or public-web metadata collection
- Today is about design judgment, not product loyalty

Question for every case: what is the artifact, what is the failure mode, and what must the economist still verify?

Three Research Patterns

Pattern	Representative task	Main risk if you use the wrong workflow
Reusable skill	Referee-style paper review	Generic comments, hallucinated quotes, missing quality gate
Empirical pipeline	MEPS harmonization and figures	Wrong crosswalk, dropped survey weights, silent validation failure
Public-web dataset	Econometric Society fellows metadata	Broken source discipline, missing provenance, hidden missingness

All three use agents, but the harness is different.

What Remains Human Across All Three

You still own

- the research question
- the coding rules
- the validation target
- the interpretation
- the decision to trust or reject the output

The agent accelerates

- repetitive execution
- file handling
- promptable decomposition
- re-runs after fixes
- draft diagnostics and plots

Lec. 9 still governs this lecture: specification and validation are not optional just because the execution is faster.